WAVE 01 OF 9 · PROJECT SKELETON

Foundation

Create the runnable project skeleton: package manifest, strict TypeScript, configuration and a health endpoint.

1 task | Starts the plan | Unlocks Wave 02

01 02 03 04 05 06 07 08 09

Why this wave exists

Every later wave needs a compiling project with a config module and a server. Nothing can be tested until this exists.

How it fits

The diagram shows this wave's tasks, the earlier-wave tasks they depend on, and the later-wave tasks that depend on them.



Hand-off. Wave 2 receives the npm scripts (typecheck, build, test) and the `src/config.ts` contract.

Tasks

Scaffolding Node TypeScript project

`scaffold-project`

Depends on: nothing

Create the Node.js and TypeScript project in the workspace root: package.json with scripts (dev, build, start, typecheck, test), a strict tsconfig.json, a config module that reads environment variables with defaults, and a minimal Express server exposing GET /health.

`package.json · tsconfig.json · src/config.ts · src/server.ts · src/main.ts`

- [ ] npm install, npm run typecheck, npm run build and npm test all succeed (a health test is enough).
- [ ] npm start serves GET /health with status 200.
- [ ] Config defaults: COUNTRY_CODE=PT, PORT=3000, PRODUCT_CONTEXT_BASE_URL and AVAILABILITY_BASE_URL=http://127.0.0.1:8080, DB_PATH=./data/processor.db, CONCURRENCY=8, CHUNK_SIZE=500, RETRY_MAX_ATTEMPTS=5, RETRY_BASE_MS=200, RETRY_MAX_MS=5000, LEASE_TTL_MS=30000.

Risks and decisions

- The stack is fixed: Express, better-sqlite3 and Vitest. Changing it later means rework in waves 2 to 6.
- better-sqlite3 needs a native build on Windows. Install it in this wave so any problem shows up early.

Verification

```
npm install
npm run typecheck
npm run build
npm test
npm start
curl http://127.0.0.1:3000/health
```

Done when

1. Every acceptance box is checked and the verification commands pass.
2. The session todos for this wave are marked done.
3. The final message includes a hand-off note for the next wave.

Agent brief for this wave: AGENT.md in the same folder.